

NanoGT Match Report: Pompe disease

Disease: Pompe disease (ORPHA:365)

Primary gene: GAA

Gene CDS: 2856 bp

Inheritance: Autosomal recessive

Target tissues scored: muscle, heart

Interpretation

- At least one high-confidence precedent was found, but this is still a precedent match rather than a clinical-trial recommendation.
- Main review flags: Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints; Vector does not naturally cover all annotated disease tissues: heart, muscle; No direct tissue overlap; treat this as weak precedent unless route or modality is changed.

Disease Mechanism Evidence

Molecular mechanism: loss of function

Mechanistic detail: Acid alpha-glucosidase lysosomal enzyme deficiency

Gene-addition compatibility: conditional

Preferred modality class: gene addition or secreted enzyme rescue

Evidence level/status: direct / source_linked_needs_review

Evidence summary: GAA deficiency supports replacement logic but skeletal and cardiac delivery needs broad tissue correction

Evidence source: [OMIM GAA gene entry](#)

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
 - Modality coverage is limited mainly to AAV and integrating ex vivo HSC vector precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
 - Endpoint risk: muscle/cardiac diseases may need functional, respiratory, imaging, or cardiac endpoints that progress slowly and vary by age/stage.
 - Endpoint risk: multi-system disease may need a hierarchy of primary and secondary endpoints; one tissue response may not equal whole-disease benefit.
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Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Libmeldy	LV	8.4/10	High	approved
2	ABO-101	AAV9	8.0/10	High	phase1/2
3	AVR-RD-01	LV	8.0/10	High	phase1/2
4	RGX-121	AAV9	8.0/10	High	phase3
5	ST-920	AAV2/6	7.9/10	High	phase1/2

Match #1: Libmeldy

Precedent disease: Metachromatic leukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 8.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	2.00	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.60	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	8.38	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 2856bp / cargo 8000bp (36% utilized)
- No tissue overlap (disease: ['muscle', 'heart'], vector: ['hematopoietic'])
- Both lysosomal enzymes — cross-correction via M6P receptor pathway likely
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: leukodystrophy
- Disease mechanism: loss of function — Acid alpha-glucosidase lysosomal enzyme deficiency
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: GAA deficiency supports replacement logic but skeletal and cardiac delivery needs broad tissue correction
- Mechanism source: OMIM GAA gene entry (<https://omim.org/entry/606800>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden

- Wide therapeutic window — adult or chronic onset; GT can be administered at multiple time-points; irreversible damage has not yet occurred at typical diagnosis age
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: moderate immune surveillance — standard immunosuppression protocols typically sufficient
- Promoter availability: MHCK7, CK8, Desmin — validated in Elevidys (DMD) and SMA programs
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera
- ORGANELLE TARGETING: COMPATIBLE — GAA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: heart, muscle
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable

Match #2: ABO-101

Precedent disease: Mucopolysaccharidosis type IIIB

Vector: AAV9

Tissue target: CNS

Composite score: 8.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	2.00	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.60	1.0	Immunological protection of target tissue

Dimension	Score	Max	What it measures
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	8.00	10.0	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 2856bp / cargo 4700bp (61% utilized)
- Vector tropism overlap: heart, muscle
- Both lysosomal enzymes — cross-correction via M6P receptor pathway likely
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: lysosomal storage
- Disease mechanism: loss of function — Acid alpha-glucosidase lysosomal enzyme deficiency
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: GAA deficiency supports replacement logic but skeletal and cardiac delivery needs broad tissue correction
- Mechanism source: OMIM GAA gene entry (<https://omim.org/entry/606800>)
- Approval status: phase1/2
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Wide therapeutic window — adult or chronic onset; GT can be administered at multiple time-points; irreversible damage has not yet occurred at typical diagnosis age
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: moderate immune surveillance — standard immunosuppression protocols typically sufficient
- Promoter availability: MHCK7, CK8, Desmin — validated in Elevidys (DMD) and SMA programs
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera
- ORGANELLE TARGETING: COMPATIBLE — GAA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #3: AVR-RD-01

Precedent disease: Fabry disease

Vector: LV

Tissue target: hematopoietic

Composite score: 8.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	2.00	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.60	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	8.00	10.0	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 2856bp / cargo 8000bp (36% utilized)
- No tissue overlap (disease: ['muscle', 'heart'], vector: ['hematopoietic'])
- Both lysosomal enzymes — cross-correction via M6P receptor pathway likely
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal storage
- Disease mechanism: loss of function — Acid alpha-glucosidase lysosomal enzyme deficiency
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: GAA deficiency supports replacement logic but skeletal and cardiac delivery needs broad tissue correction
- Mechanism source: OMIM GAA gene entry (<https://omim.org/entry/606800>)
- Approval status: phase1/2
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Wide therapeutic window — adult or chronic onset; GT can be administered at multiple time-points; irreversible damage has not yet occurred at typical diagnosis age
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: moderate immune surveillance — standard immunosuppression protocols typically sufficient
- Promoter availability: MHCK7, CK8, Desmin — validated in Elevidys (DMD) and SMA programs

- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera
- ORGANELLE TARGETING: COMPATIBLE — GAA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: heart, muscle
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable

Match #4: RGX-121

Precedent disease: Mucopolysaccharidosis type II

Vector: AAV9

Tissue target: CNS/liver

Composite score: 8.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.80	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	2.00	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.60	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	8.00	10.0	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 2856bp / cargo 4700bp (61% utilized)
- Vector tropism overlap: heart, muscle
- Both lysosomal enzymes — cross-correction via M6P receptor pathway likely
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal_storage
- Disease mechanism: loss of function — Acid alpha-glucosidase lysosomal enzyme deficiency
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: GAA deficiency supports replacement logic but skeletal and cardiac delivery needs broad tissue correction
- Mechanism source: OMIM GAA gene entry (<https://omim.org/entry/606800>)
- Approval status: phase3
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Wide therapeutic window — adult or chronic onset; GT can be administered at multiple time-points; irreversible damage has not yet occurred at typical diagnosis age
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: moderate immune surveillance — standard immunosuppression protocols typically sufficient
- Promoter availability: MHCK7, CK8, Desmin — validated in Elevidys (DMD) and SMA programs
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera
- ORGANELLE TARGETING: COMPATIBLE — GAA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #5: ST-920

Precedent disease: Fabry disease

Vector: AAV2/6

Tissue target: liver

Composite score: 7.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class

Dimension	Score	Max	What it measures
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	1.50	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	2.00	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.60	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.86	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 2856bp / cargo 4700bp (61% utilized)
- Vector tropism overlaps muscle, but precedent target is liver
- Both lysosomal enzymes — cross-correction via M6P receptor pathway likely
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal storage
- Disease mechanism: loss of function — Acid alpha-glucosidase lysosomal enzyme deficiency
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: GAA deficiency supports replacement logic but skeletal and cardiac delivery needs broad tissue correction
- Mechanism source: OMIM GAA gene entry (<https://omim.org/entry/606800>)
- Approval status: phase1/2
- Vector immunogenicity (AAV2/6): moderate (~17%) — significant proportion may require pre-screening or exclusion
- Wide therapeutic window — adult or chronic onset; GT can be administered at multiple time-points; irreversible damage has not yet occurred at typical diagnosis age
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: moderate immune surveillance — standard immunosuppression protocols typically sufficient
- Promoter availability: MHCK7, CK8, Desmin — validated in Elevidys (DMD) and SMA programs
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera
- ORGANELLE TARGETING: COMPATIBLE — GAA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Dual critical target tissues; confirm whether one route can plausibly address both clinical endpoints
- Vector does not naturally cover all annotated disease tissues: heart
- Only partial tissue match; verify target-cell transduction and delivery route manually
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone